INCH-POUND
MIL-DTL-24643/26E
22 August 2002
SUPERSEDING
MIL-C-24643/26D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPES LSDPS, LSFPS, LSTPS AND LS7PS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First -	Copper conductor, nickel coated (see table I for size and stranding).
Second -	Extruded silicone rubber insulation (see table I for dimensions).
Third -	Glass braid (see table I for dimensions). Standard identification code applied by method 4.
Fourth -	Transparent braid covering. The specified number of pairs (see table I) cabled together with a lay not greater than 10 inches. Fillers saturated with non-charring silicone based compound to form a firm, well-rounded assembly.
Fifth -	A binder tape applied helically with overlap.
Sixth -	Silicone rubber jacket colored orange or red (see table I for thickness).
Seventh -	Braided metal armor.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/26	Type and size	Conductors				Cable jacket thickness min avg (inch)	Overall diameter maximum (inch)	Cold bending mandrel dia max (inches)	Conductor resistance (ohms)
		No. of CDRs	AWG size and stranding	Diameter over insulation minimum (inch)	Diameter over braid covering maximum (inch)				
-01AN	LSDPS-3	2	16 (class B)	0.100	0.125	0.038	0.455	6.0	5.39
-02AN	LSDPS-4	2	14 (class B)	0.116	0.141	0.038	0.489	6.0	3.38
-03AN	LSDPS-6	2	12 (class B)	0.143	0.170	0.050	0.585	7.5	2.13
-04AN	LSDPS-9	2	10 (class B)	0.166	0.193	0.050	0.628	7.5	1.34
-05AN	LSDPS-14	2	9 (class B)	0.200	0.230	0.063	0.730	8.0	1.060
-06AN	LSTPS-3	3	16 (class B)	0.100	0.125	0.038	0.475	6.5	5.39
-07AN	LSTPS-4	3	14 (class B)	0.116	0.141	0.050	0.553	7.0	3.38
-08AN	LSTPS-6	3	12 (class B)	0.143	0.170	0.050	0.620	7.5	2.13
-09AN	LSTPS-9	3	10 (class B)	0.166	0.193	0.050	0.657	7.5	1.34
-10AN	LSTPS-14	3	9 (class B)	0.200	0.230	0.063	0.751	9.5	1.060
-11AN	LSTPS-23	3	7 (class C)	0.257	0.291	0.063	0.866	10.5	0.643
-12AN	LSTPS-30	3	5 (class B)	0.310	0.348	0.075	0.989	12.0	0.403
-13AN	LSFPS-14	4	9 (class B)	0.200	0.230	0.063	0.815	10.0	1.060
-14AN	LS7PS-6	7	12 (class B)	0.143	0.170	0.063	0.775	9.5	2.13
-15AN	LS7PS-14	7	9 (class B)	0.200	0.230	0.063	0.986	11.0	1.060

EXAMINATION AND TESTS:

Basic electrical: Requirements:

- Conductor resistance - ohms/1000 feet at 25°C, maximum ..... (see table I)
- Voltage withstand - volts, root mean square, minimum 3000
- Insulation resistance - megohms -1000 feet, minimum
  - Conductor to conductor 500
  - Conductor continuity ..... No failure

Group A:

- Visual and dimensional ..... No failure
- The nickel coating on the conductors of the completed cable shall not be nicked, scraped or abraded so as to expose the copper conductor ..... No failure
- Watertightness - see MIL-DTL-24643 for limits of water leakage ..... No failure

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (When tested at 200°C) 50

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Gas flame - 1 hour at 230/130 volt root mean square ..... No failure
Drip - 95 ± 1°C ..... Zero
Cold bending, cable - (see table I for mandrel diameter) ..... No damage

Physicals (unaged)

Insulation

Tensile strength - lb/in^2, minimum 750
Elongation - percent, minimum 125

Jacket (cable)

Tensile strength - lb/in^2, minimum 800
Elongation - percent, minimum 250
Durometer (type A) hardness, minimum 60

Group C:

Physicals - cable aging (260 ± 10°C)

Jacket

Tensile strength - lb/in^2, minimum 600
Elongation - percent, minimum 150
Cable sealant removability ..... No failure

Group D:

Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Gas flame (3 hours) at 120/70 volts RMS ..... No failure
Aging and compatibility (cable) (125 ± 5°C) ..... No failure
For this test, omit all references to watertightness and fillers.
Armor - conformance to material, construction and coverage ..... No failure
Specific gravity of extruded silicone insulation, maximum 1.55

Halogen content - percent, maximum

Jacket 0.2
Fillers 0.2

Smoke index, maximum

Jacket 35
Fillers 45
Insulation 35

Toxicity index, maximum

Jacket 5
Fillers 5
Insulation 1.5

UNIT ORDERING LENGTHS:

All types and sizes 100 feet minimum.

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Custodians:
Army - MI
Navy - SH

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Preparing Activity:
Navy - SH
(Project 6145-2308-024)

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